

Where to spend the health intervention budget

A segmentation of 1,337 members identifying the highest-yield target for cost-avoidance intervention.

Provenance. The source file is the widely circulated Medical Cost Personal Datasets teaching set, which is simulated rather than drawn from real claims. The findings are real properties of the data and the methods are the ones that would be used on a live book, but no conclusion here describes an actual insurer or population.

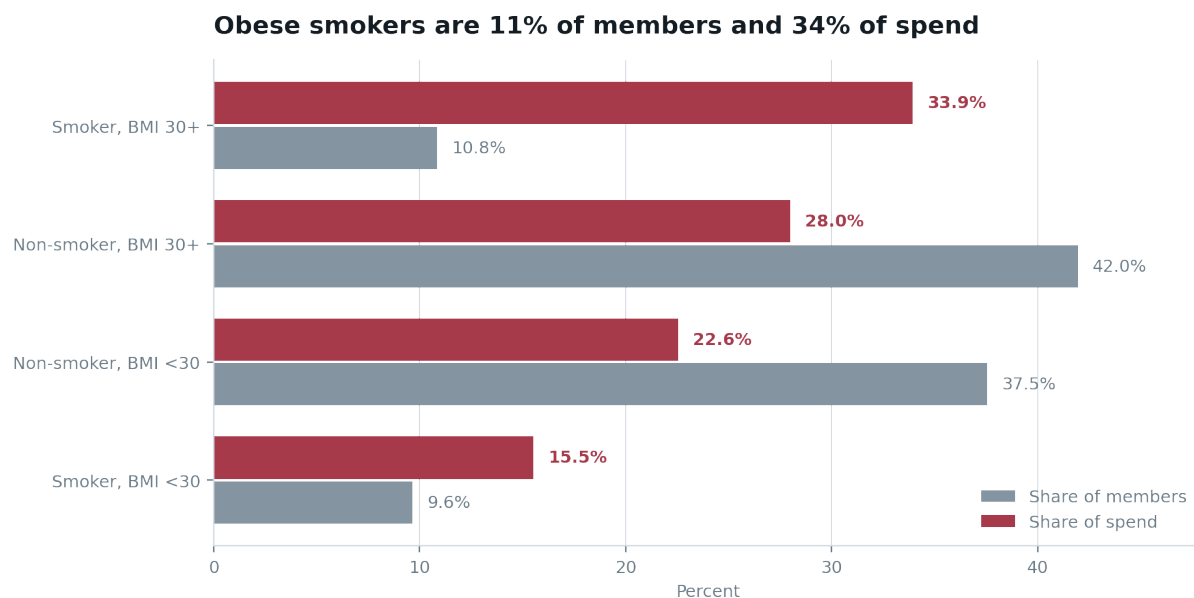
The recommendation

Point the health intervention budget at smokers with a BMI of 30 or above. There are 145 of them in a book of 1,337 members — 10.8% of the population — and they account for \$6.03M, or 33.9% of total spend. Their mean annual cost is \$41,558 against \$7,977 for members carrying neither risk factor.

Smoking cessation is the lever, not weight management. That is the counterintuitive part of this analysis and it is where the money is.

Why the two risk factors cannot be treated separately

Obesity costs a non-smoker \$879 a year. It costs a smoker \$20,195 — a 23-fold difference in the same risk factor. Neither variable on its own identifies where to intervene; only the intersection does.



Population and cost are inverted. Only the obese-smoker segment consumes materially more than its share of the book.

What each option is worth

The ceiling column assumes every member in the segment converts and carries no residual risk. It is an upper bound, not a forecast. The 10% column is the figure to take into a business case — generous but defensible for cessation programmes, optimistic for weight programmes.

Lever	Members	Ceiling	% of book	At 10%
Smoking cessation — obese smokers	145	\$4.74M	26.7%	\$474k
Smoking cessation — non-obese smokers	129	\$1.73M	9.7%	\$173k
Weight programme — obese non-smokers	561	\$0.49M	2.8%	\$49k

The third row is the trap. Obese non-smokers are the single largest group in the book at 561 members — 42% of the population — and a general wellness programme would naturally aim there. The entire theoretical prize is \$493k, or \$879 a head. The obese-smoker segment is a quarter the size and worth 9.6 times as much.

Three things that change the recommendation

Target the young, not the old

The smoker-to-non-smoker cost gap is close to constant across the age range: \$23,091 at 18-29, \$23,934 at 30-39, \$23,471 at 40-49, \$25,317 at 50-64. Because the annual gap does not widen, the younger cohort is worth more in total — the same saving repeats over more remaining years. Outreach aimed at older smokers on the assumption that they are more expensive to leave untreated is not supported here.

Run it in the southeast first

Obese-smoker prevalence is 15.9% in the southeast against roughly 9% across the other three regions, and that segment accounts for 45.7% of all southeast spend. Geography is not a separate lever; it is the delivery sequence.

Do not price anything on the BMI 30 boundary

The apparent \$20,195 step at BMI 30 is an artifact of how this dataset was generated, not a physiological threshold. The full argument is in the data quality report. Any eligibility rule or saving estimate built on that cutoff inherits the artifact.

What to do next

- Commission a cessation programme scoped to the obese-smoker segment, delivered southeast-first, with explicit outreach to the 18–29 cohort.
- Instrument the data before the next cycle. A linear model on the six available inputs reaches $R^2 = 0.75$; the missing quarter is concentrated in members with no visible risk factors. Diagnosis codes, claim dates and member IDs would close it.
- Verify smoking status. The entire targeting list rests on a self-reported binary field, and tobacco use is systematically under-declared in insurance settings.
- Do not fund a general weight programme on cost-avoidance grounds. It may be justified on health outcomes; it is not justified by this data on spend.

These are not causal effects. The dataset has no time dimension, so nothing here can demonstrate that removing a risk factor causes a cost reduction. Every saving figure is an association-based ceiling: it moves a segment onto the observed mean of the segment it would belong to without that risk factor, assuming complete conversion and no residual risk.